==========SQL==========

CREATE TABLE groceries (id INTEGER PRIMARY KEY, name TEXT, quantity INTEGER, aisle INTEGER);

Insert into employee_table (employee_id, employee_name, client_name, department_name, age) values (1, “Ariel jay Fuentes”, “HEB”, “IT”, 30);

Select * From employee_table WHERE client_name=”HEB” and department_name=”IT”; **AND**
Select * From employee_table WHERE department_name =”HR” and department_name=”IT”; **OR**
Select * From employee_table WHERE NOT department_name=”IT”; N**OT**
Select * From employee_table WHERE department_name IS NULL; **IS NULL/IS NOT NULL**
Select * From employee_table WHERE NOT department_name=”IT” order by employee_ID ASC; **ORDER BY … ASC|DESC**
Select DISTINCT * From employee_table WHERE client_name=”HEB” and department_name=”IT”; // only one result **DISTINCT**
Select * from employee_table limit 10; //limit number of rows to display **LIMIT**
Select max(age) from employee_table ; **MAX**
Select min(age) from employee_table ; **MIN**

Select max(salary) from employee_table ; //**select the highest salary**
Select * from employee where salary in(Select Max(salary) from employee); //get the employee with highest salary
Select max(salary) **from employee where salary NOT in(Select Max(salary) from employee); //get the 2nd highest salary**
Select e.name, e.salary, d.departmentName from Employee e inner join Department d on e.departmentId=d.departmentId // get the employee details together with department name found in department table.
Select e.name, e.salary, d.departmenName from Employee e inner join Department d on e.departmentId=d.departmentId where salary in(
Select max(salary) from employee
) // employee with highest salary together with other details such as departmentName

Select e.salary, d.departmentName from Employee e inner join Department d on e.departmentId=d.departmentId where salary in (select max (salary) from employee group by departmentId))
// return highest salary for each department

Select name, salary from employee where salary > all (select salary Employee e inner join Department d on d.departmentId=e.departmentId where d.departmentName=”IT”)
// select the names of employee whose salary is greater than any employee in IT Department.

Select * from (select d.departmentName as cn , e.departmentId as dId, avg(e.salary) as aveSalary from Employee e. inner join Department d on d.departmentId=e.departmentId group by d.departmentName) as tb, employee ne where tb.dId=ne.departmentId and ne.salary > tb.aveSalary;

//select employees whose salary is greater than departments average salary

Update employee_table set department_name= “IT ” where employee_id=1;
Delete from employee_table where employee_id=1;

COALESCE function that allows you`SELECT COALESCE(NULL, 0); -- 0 SELECT COALESCE(NULL, NULL); -- NULL;` to substitute NULL values.

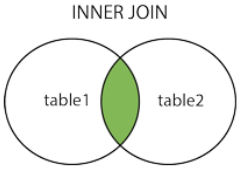
Queries Types

1. Data Definition Queries – Create, Alter
2. Data Manipulation Queries - Insert, Update, Delete

JOINS- Used to combine rows from two or more tables

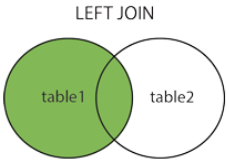
INNER JOINS - Selects records that have matching values in both tables

SELECT column_name(s) FROM table1 INNER JOIN table2 ON table1.column_name = table2.column_name;

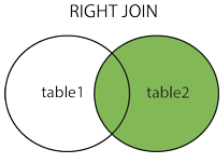


LEFT JOIN -return all records on the left table and match the record from the right table. Right is null if there's no match on left.

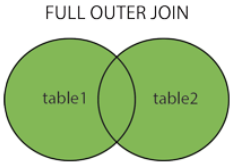
SELECT column_name(s)FROM table1 LEFT JOIN table2 ON table1.column_name = table2.column_name;



RIGHT JOIN-return all records on the right table and match the record from the left table. left is null if there's no match on right.



FULL JOINreturns all records when there is a match in left (table1) or right (table2) table records.



SELF JOIN –table is joined with itself.

SELECT column_name(s) FROM table1 T1, table1 T2 WHERE condition;